

CHICAGO CALUMET HARBOR, IL, USA

WMO: 997255

Lat: 41.730N

Lon: 87.538W

Elev: 179

StdP: 99.19

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-14.6	N/A	N/A	N/A	N/A	N/A	N/A	11.2	-3.3	10.2	-0.7	4.2	250	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	32.0	N/A	30.5	N/A	28.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	9.9	8.4	-22.5	33.6	4.0	1.9	-25.4	35.0	-27.8	36.1	-30.1	37.2	-33.0	38.6		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-3.6	-2.8	2.6	9.2	14.9	20.4	23.3	22.8	19.0	12.4	5.5	-0.3
	DBStd	10.69	6.47	5.71	5.17	4.83	5.13	4.27	3.43	2.61	3.88	4.70	5.42	5.69
	HDD10.0	1601	423	358	238	71	12	0	0	0	0	27	152	320
	HDD18.3	3392	681	591	488	276	134	28	3	1	37	192	384	577
	CDD10.0	1732	1	1	8	49	163	313	411	397	269	102	18	2
	CDD18.3	483	0	0	1	3	28	91	156	139	56	9	0	0
	CDH23.3	3181	0	0	1	26	222	644	1143	776	322	47	0	0
	CDH26.7	945	0	0	0	2	63	191	390	204	87	8	0	0
Wind	WSAvg	4.0	4.2	4.2	5.0	4.9	4.1	3.8	3.4	3.3	3.6	3.8	4.1	4.1
Precipitation	PrecAvg	898	53	46	57	83	105	99	98	99	71	75	65	49
	PrecMax	1223	129	151	125	176	191	278	247	224	239	216	170	108
	PrecMin	521	15	13	14	34	17	24	30	22	19	22	10	8
	PrecStd	160	31	28	29	35	43	52	47	50	48	48	41	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	14.5	22.0	26.6	31.1	33.0	35.3	32.6	32.0	27.8	20.8	15.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	10.7	10.1	17.7	23.6	28.4	30.8	32.5	31.0	29.1	24.4	17.8	12.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	7.6	6.7	14.0	20.8	25.9	28.9	30.6	29.0	26.9	21.8	15.6	9.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	4.7	4.0	10.0	17.9	23.1	27.0	28.9	27.3	24.8	19.5	13.3	7.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.8	6.5	6.5	7.8	8.1	7.4	6.5	5.9	6.6	7.1	6.8	6.0
		MCDBR	9.6	9.7	12.9	13.4	12.1	10.3	9.3	9.0	9.7	10.1	9.4	8.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.306	0.329	0.353	0.409	0.438	0.443	0.455	0.452	0.422	0.365	0.323	0.306	
	taud	2.438	2.359	2.368	2.238	2.211	2.248	2.214	2.219	2.264	2.425	2.496	2.483	
	Ebn at Noon	844	873	893	861	842	836	821	811	808	823	818	814	
	Edh at Noon	77	98	109	134	141	136	139	135	120	91	73	69	
All-Sky Solar Radiation	RadAvg	1.59	2.44	3.58	4.61	5.25	6.05	6.10	5.30	4.38	2.89	1.83	1.33	
	RadStd	0.17	0.32	0.27	0.38	0.45	0.39	0.29	0.31	0.37	0.30	0.19	0.15	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page